

IDEALS AND LOCALISATION

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ABSTRACT. A small expository note reviewing ideal correspondence theorems in Commutative Algebra. Then, we ask the question of when maximal ideals pull back to maximal ideals, giving counterexamples, a positive result, and a geometric interpretation. This is all well known.

A key thing that is important to understand in the world of commutative algebra is how ideals interact with ‘subobjects’. The first result in this direction is the *correspondence theorem*.

Theorem (The Correspondence Theorem). *Let A be a commutative ring, $I \triangleleft A$ an ideal. Let $\pi : A \rightarrow A/I$ denote the canonical projection map. Then we have the following bijections:*

$$\begin{aligned} \{\text{Ideals } J \triangleleft A/I\} & \xrightleftharpoons[\pi(J)]{\pi^{-1}(J/I)} \{\text{Ideals } I \subset J \triangleleft A\} \\ \{\text{Prime Ideals } P \triangleleft A/I\} & \xrightleftharpoons[\pi(P)]{\pi^{-1}(P/I)} \{\text{Prime Ideals } I \subset P \triangleleft A\} \\ \{\text{Maximal Ideals } m \triangleleft A/I\} & \xrightleftharpoons[\pi(m)]{\pi^{-1}(m/I)} \{\text{Maximal Ideals } I \subset m \triangleleft A\} \end{aligned}$$

When I first came across this result, I remember the first bijection being proved in full, then the second and third being left as exercises. I proved the second bijection, then due to a lack of vigilance I didn’t prove the third, thinking it was clear. I have since amended that, but it lead to some misunderstandings, which I would like to share, these arise when considering other ‘correspondence theorems’.

Theorem. *Let $\phi : A \rightarrow B$ be a morphism of rings. Then if we have an ideal $I \triangleleft B$, then $\phi^{-1}(I)$ is an ideal of A . Furthermore, if $P \triangleleft B$ is a prime ideal, then $\phi^{-1}(P)$ is a prime ideal of A .*

Counterexample. The above theorem fails for maximal ideals! Consider the inclusion $\iota : \mathbb{Z} \rightarrow \mathbb{Q}$. The maximal $(0) \triangleleft \mathbb{Q}$ is pulled back to (0) in $\mathbb{Z}$, which is not maximal.

We are also interested in how prime ideals interact with localisation due to the geometric interpretation of localisation. Most important are the following flavours of localisation.

Definition. Let A be a commutative ring, $f \in A$, $P \triangleleft A$ a prime ideal.

- (a) If $S = \{1, f, f^2, f^3, \dots\}$, then we denote $S^{-1}A$ by A_f .
- (b) If $S = A \setminus P$, then we denote $S^{-1}A$ by A_P .

Moreover, we call the natural map $\varphi : A \rightarrow S^{-1}A$, $\varphi(a) = a/1$ the *localisation homomorphism*

Relating ideals of $S^{-1}A$ to A is, in some sense, the same as understanding how they behave under the localisation homomorphism (this was the same strategy as before). In order to do this, we introduce some notation.

Definition. Let A be a commutative ring, $S^{-1}A$ the localisation at some multiplicative subset S , and φ the localisation homomorphism. Let $I \triangleleft R$ be an ideal. Define the *extension of I along φ* , I^e as the ideal $I^e = S^{-1}A \cdot \varphi(I)$. Given an ideal $J \triangleleft S^{-1}A$, define the *contraction of J along φ* , J^c as the ideal $f^{-1}J$.

It is easily verified that $I^e = S^{-1}I = \{a/s \mid a \in I, s \in S\}$. The most important correspondence theorem in this case is the following:

Theorem. Let A be a commutative ring, $S \subset A$ be a multiplicative set. Then contraction and extension along φ induce the following bijection:

$$\{\text{Prime Ideals } Q \triangleleft S^{-1}A\} \xleftrightarrow{\sim} \{\text{Prime ideals } P \triangleleft A \mid P \cap S = \emptyset\}$$

Remark. Specialising to our most important flavours of localisation, we have the following bijection:

$$\begin{aligned} \{\text{Prime Ideals } Q \triangleleft A_P\} &\xleftrightarrow{\sim} \{\text{Prime Ideals } Q \subset P\} \\ \{\text{Prime Ideals } Q \triangleleft A_f\} &\xleftrightarrow{\sim} \{\text{Prime Ideals } P \triangleleft A \mid f \notin P\} \end{aligned}$$

Again, the above does *not* hold for maximal ideals!

Counterexample. Consider $A = \mathbb{C}[x, y]$ and the prime ideal $P = (x^2 + y^2 - 1)$. It is well known that A_P is a local ring with maximal ideal P^e . However, $P^{ec} = P$, which is not maximal.

Counterexample. Consider A_P where $A = \mathbb{C}[x]$ and $P = (x)$. If we further localise at $\{1, x, x^2, \dots\}$, we are left with $\mathbb{C}(x)$, which is a field. The only maximal ideal is (0) , which pulls back to (0) . This is not maximal in A_P - it is a local ring with maximal ideal (x) .

There is a class of rings where localisations of the form A_f do indeed satisfy the property that maximal ideals contract to maximal ideals, these are the finitely-generated k -algebras, for some field k . This has a very neat geometric interpretation, which is what we will present. For this, we require the following result.

Proposition (Zariski's Lemma). *Let k be a field, every maximal ideal of $k[x_1, \dots, x_n]$ has residue field a finite extension of k . In other words, any field extension of k that is finitely-generated as a k -algebra is a finite field extension.*

Proposition. *Let A be a finitely generated k -algebra. Let $f \in A$ be a non-zero element. Let φ be the localisation homomorphism $\varphi : A \rightarrow A_f$. Let $m \triangleleft A_f$ be maximal. Then $m^c = \varphi^{-1}(m)$ is a maximal ideal of A .*

Proof. By Zariski's Lemma, it suffices to show that A/m^c is a finite-dimensional vector space over k . As m is maximal, it is prime. Thus m^c is prime, and A/m^c is an integral domain. Therefore, A/m^c injects into $(A/m^c)_f$. Moreover, $(A/m^c)_f \cong A_f/m$ as k -vector spaces, and A_f/m is a finite-dimensional vector space k -vector space - A_f is a finitely generated k -algebra, as $A_f \cong A[t]/(tf - 1)$. Therefore, A/m^c is a k -vector space which injects into a finite-dimensional vector space. Thus it is finite-dimensional, and by Zariski's Lemma, m^c is maximal. $\square$

This has a very nice geometric interpretation. From here, I freely use geometric language: see Vakil's *Foundations of Algebraic Geometry*. This is slightly sketchy, as a full exposition would lead to terrible scope creep.

Proposition ([1] Exercise 3.6.I). *Suppose that A is a finitely-generated k -algebra. Then the set of closed points of $\text{Spec } A$ is dense.*

Proof. We will show that if $f \in A$ such that $D(f)$ is nonempty, then $D(f)$ contains a closed point of $\text{Spec } A$. This implies density as for any $[p] \in \text{Spec } A$, any open neighbourhood of $[p]$ contains a closed point.

Let $f \in A$ be such that $D(f)$ is nonempty. This occurs iff $f \notin N(A)$, where $N(A)$ denotes the nilradical of A . Thus $0 \notin \{1, f, f^2, \dots\}$, so A_f is not the zero ring. Thus A_f contains a maximal ideal, which is of the form p^e , for some prime $p \triangleleft A$. By the above proposition, this is maximal in A . We can identify $\text{Spec } A_f$ with $D(f)$ as schemes, and we can identify closed points with maximal ideals in the case of affine schemes. Therefore, $D(f)$ contains a closed point, and we are done. $\square$

REFERENCES

- [1] Ravi Vakil (2025), *The Rising Sea*, Princeton University Press
- [2] Marco Schlicting (2025), *MA3G6 Lecture Notes*, Unpublished.